

KIRO WORKSHOP

2026新北市AI智慧城市黑客松競賽 實戰工作坊

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使用 Kiro 打造小遊戲！



AWS AI Stack

APPLICATIONS

Kiro

Amazon Q

AWS Transform

Amazon Connect

AWS Marketplace

開箱即用的智能助手

AI & AGENT DEVELOPMENT SOFTWARE & SERVICES

SDKS FOR AGENTS

Vertically Integrated

Nova Act

Flexible/OSS

Strands Agents

GenAI與Agent開發平台

AMAZON BEDROCK

Models

Amazon Nova

3P Models

Capabilities

Optimization

Guardrails

Customization

PartyRock (Amazon Bedrock Playground)

AgentCore

Runtime

Gateway

Memory

1P Tools

Identity

Observability

Knowledge Bases

INFRASTRUCTURE

AMAZON SAGEMAKER AI

Model building

Model training

Fine-tuning

Deployment

MLOps

Governance

AI COMPUTE

AWS Trainium

AWS Inferentia

GPUs

模型開發與基礎架構

INTERFACES & PROTOCOLS
(MCP/A2A)

DATA



Kiro - 從原型到正式上線的 Agentic AI 開發環境

AI 正在改變軟體開發

2024

ASSISTANTS

協助開發者更快速地
撰寫程式碼

2025

AGENTS

在人類參與 (Human
in the Loop) 的情況下，
端到端完成開發任務

2026

AUTONOMY

在受控範圍內，以有
限自主能力端到端完
成開發任務

Agentic 開發的未來願景

自主性

Agents 能夠自主承接並完成越來越複雜的任務

真正的協作

開發者與 Agents 協同合作，以完成更多工作

更高品質

Agents 可負責大量繁重工作，包括提升測試覆蓋率、撰寫文件、效能優化，以及修補安全漏洞

AI 開發工具面臨的挑戰

AI 工具的擴展性

AI 程式碼工具擅長處理小任務，但在複雜的項目中可能會失敗

控制有限

現有工具使得與Agent協作和管理變得困難

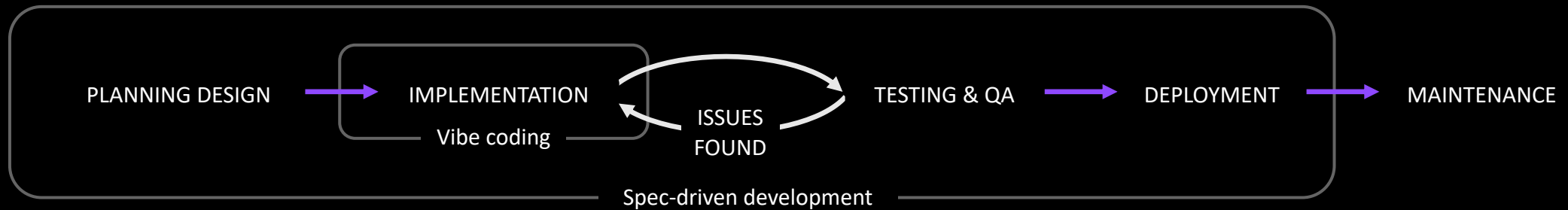
程式碼品質

在保持品質控制的同時將項目從概念驗證到生產變得越來越困難

如果能夠充分發揮與 AI Agents
協作的能力，未來的開發體驗會
是什麼樣子？

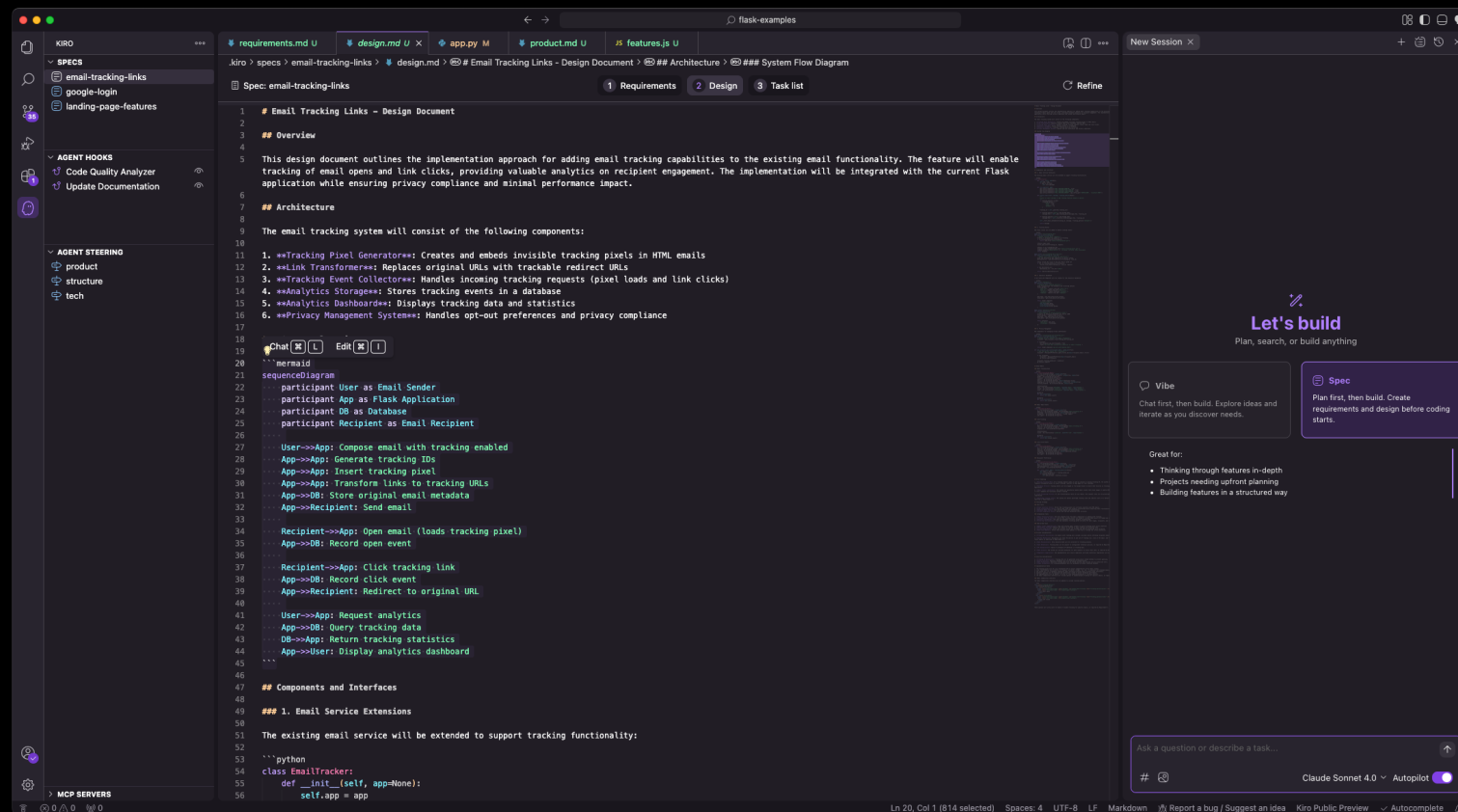
解決方案：Spec-driven Development

Vibe Coding 能快速產出 Prototype，
而傳統 SDLC（軟體開發生命週期）流程則負責交付正式產品。



Kiro 協助開發者與工程團隊運用 AI Agents，
交付高品質軟體。

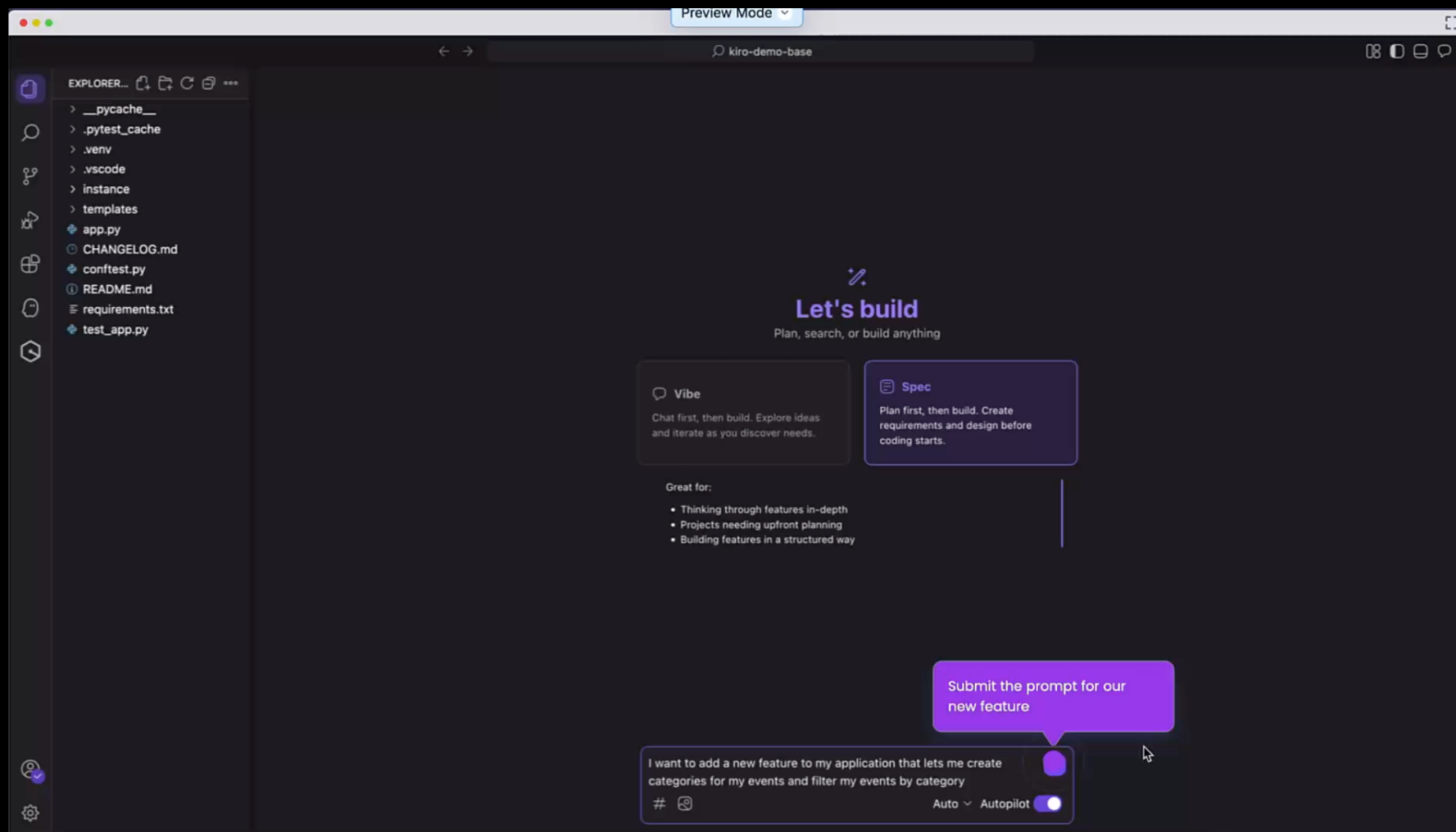
從原型到生產的 Agentic IDE



Kiro 通過 Spec-driven development 為 AI 開發引入更加**可控的結構化**工作流，從而幫助您盡力完成工作。

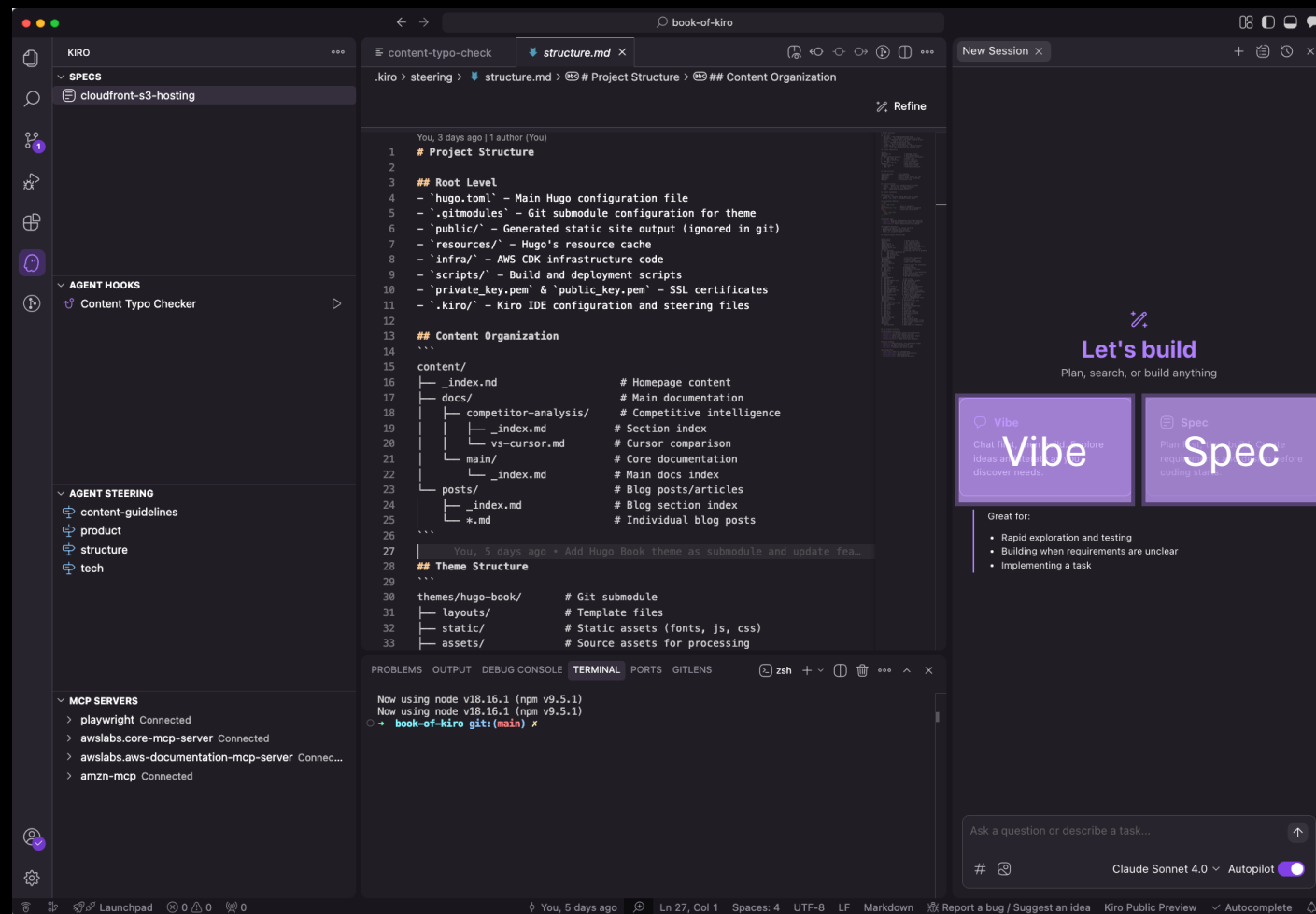
Spec-driven development

- Kiro 將您的提示詞轉化為明確的要求、系統設計和明確的任務
- Spec 使用 EARS (Easy Approach to Requirements Syntax) 來描述需求，讓需求更清晰、可實現
- Kiro Agent 在實施規範的同時保持您的控制權



構建方式：Vibe & Spec

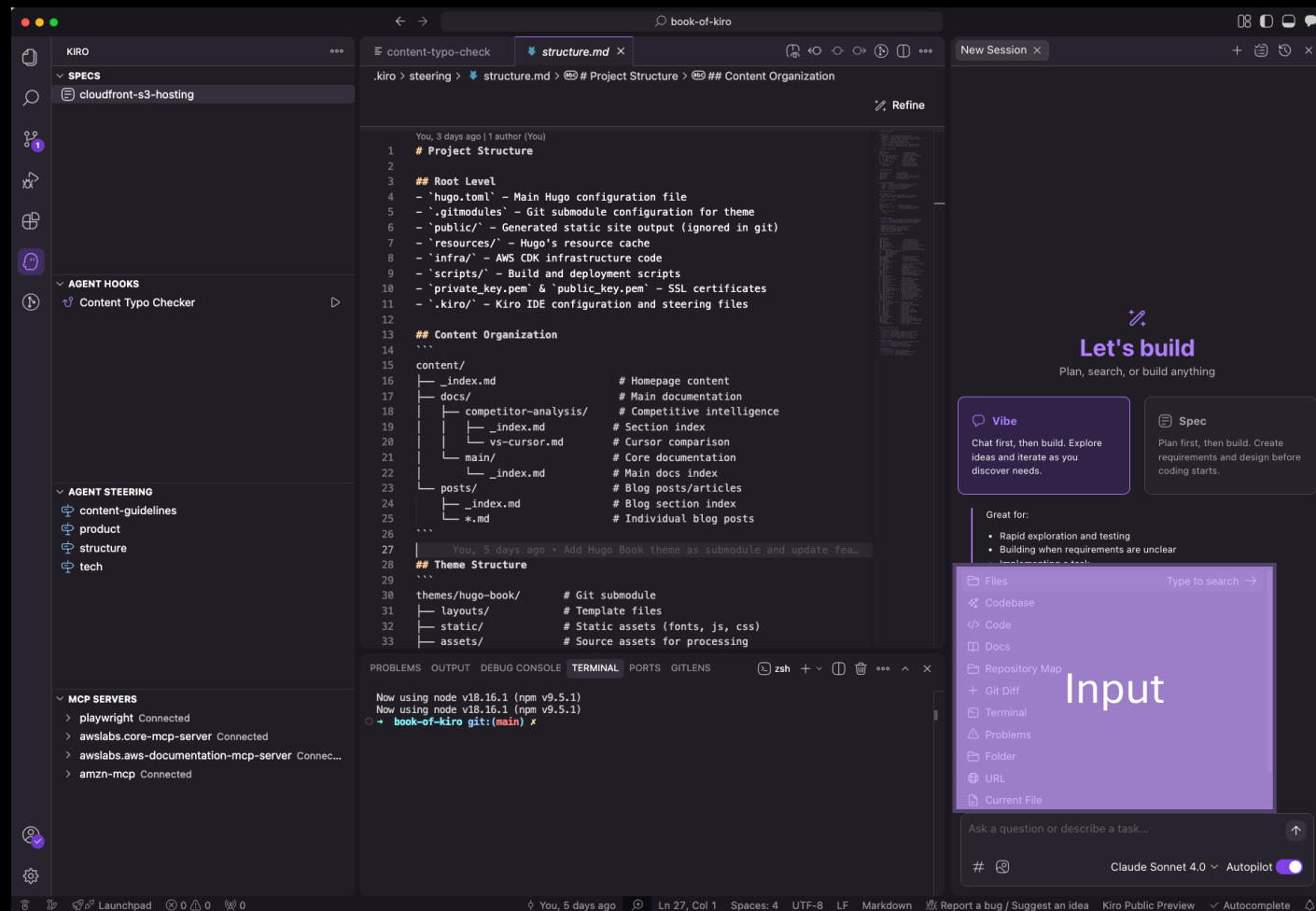
- **Vibe:** Vibe 是 Kiro 中以互動問答為核心的會話模式，專為快速提問、解釋說明以及通過更加對話化的方式構建項目而設計
- **Spec:** Spec 在 Kiro 中為您提供結構化的方法來處理複雜的開發任務，將軟體開發流程正規化。它能夠將高層次的想法轉化為詳細的實施計劃，並進行系統化執行和清晰的跟蹤



-
- The screenshot displays the Kiro IDE interface, which is a code editor with a sidebar on the left and a main workspace on the right.
- Left Sidebar:**
- Specs:** A list of specifications, including "cloudfront-s3-hosting".
 - AGENT HOOKS:** A section containing "Content Typo Checker".
 - AGENT STEERING:** A section with a list of steering rules: "content-guidelines", "product", "structure", and "tech".
 - MCP SERVERS:** A section listing connected servers: "playwright", "awslabs.core-mcp-server", "awslabs.aws-documentation-mcp-server", and "amzn-mcp".
- Main Workspace:**
- File Explorer:** Shows the project structure, including "content-typo-check" and "structure.md".
 - Code Editor:** Displays the content of "structure.md", which defines the project structure and content organization. The code includes comments and a list of files/directories.
 - Terminal:** Shows the output of the "book-of-kiro" command, indicating the current directory and the version of the tool.
- Right Panel:**
- Let's build:** A section with the text "Plan, search, or build anything".
 - Vibe:** A section with the text "Chat first, then build. Explore ideas and iterate as you discover needs."
 - Spec:** A section with the text "Plan first, then build. Create requirements and design before coding starts."
 - Great for:** A list of features: "Rapid exploration and testing", "Building when requirements are unclear", and "Implementing a task".
 - Autopilot ON/OFF:** A toggle switch at the bottom right, currently set to "ON".

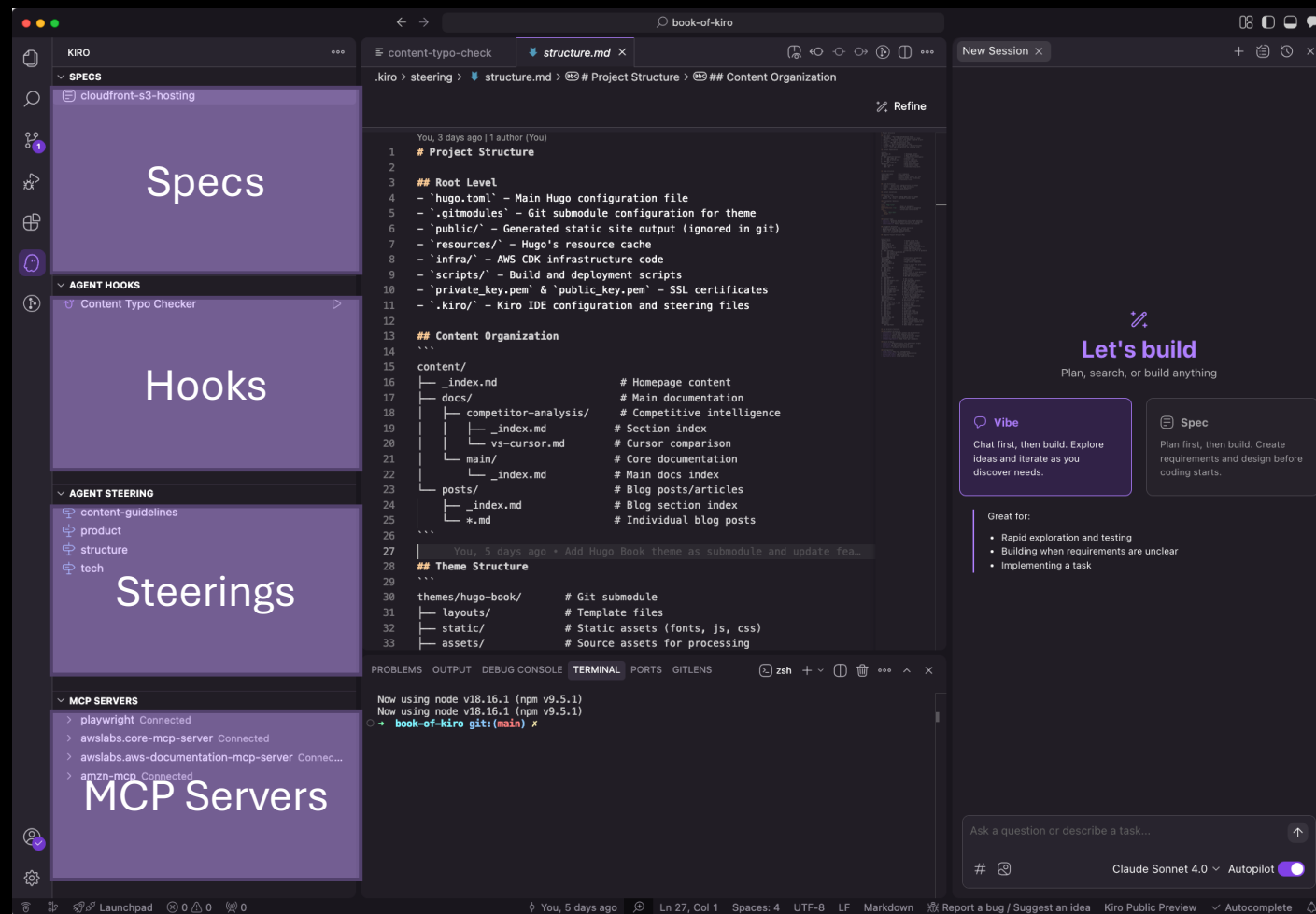
上下文 (Context)

- **Codebase**: 當前項目的程式碼索引
- **Files/Folders**: 文件或文件夾
- **Code**: 程式碼片段
- **Docs**: 常用開發文檔
- **Git Diff**: 當前的 Git Diff
- **Terminals**: 命令行中的內容
- **Problems**: 當前打開文件中被 IDE 自動檢測出來的問題
- **URL**: 給定一個 URL 的網頁內容
- **Steering**: Kiro 的 Steering 文件
- **MCP**: 調用某個 MCP



Kiro 佈局

- **Specs:** 管理程式碼生成過程中的 Specs 文件，任務有跡可循，團隊協作更方便。
- **Hooks:** 管理的 Agent Hooks 設定，讓團隊協作中的成員保持項目規範統一。
- **Steering:** 通過 markdown 文件為 Kiro 提供關於您項目的知識，指導文件確保 Kiro 始終遵循您已建立的模式、庫和標準。
- **MCP Servers:** 通過 MCP 來擴展 Kiro 的功能，為其提供更多工具和上下文資訊。



Agent hooks

- Kiro IDE 捕獲“文件建立”，“文件保存”，“文件刪除”事件（目前支援），自動將任務委派給 Agent，在後台自動執行。
- 用戶通過自然語言描述，Kiro 自動生成 Agent Hooks 的任務描述
- Agent Hooks 可以在生成文件、單元測試或優化程式碼等等方面幫助您完善 workflow

生成文件：

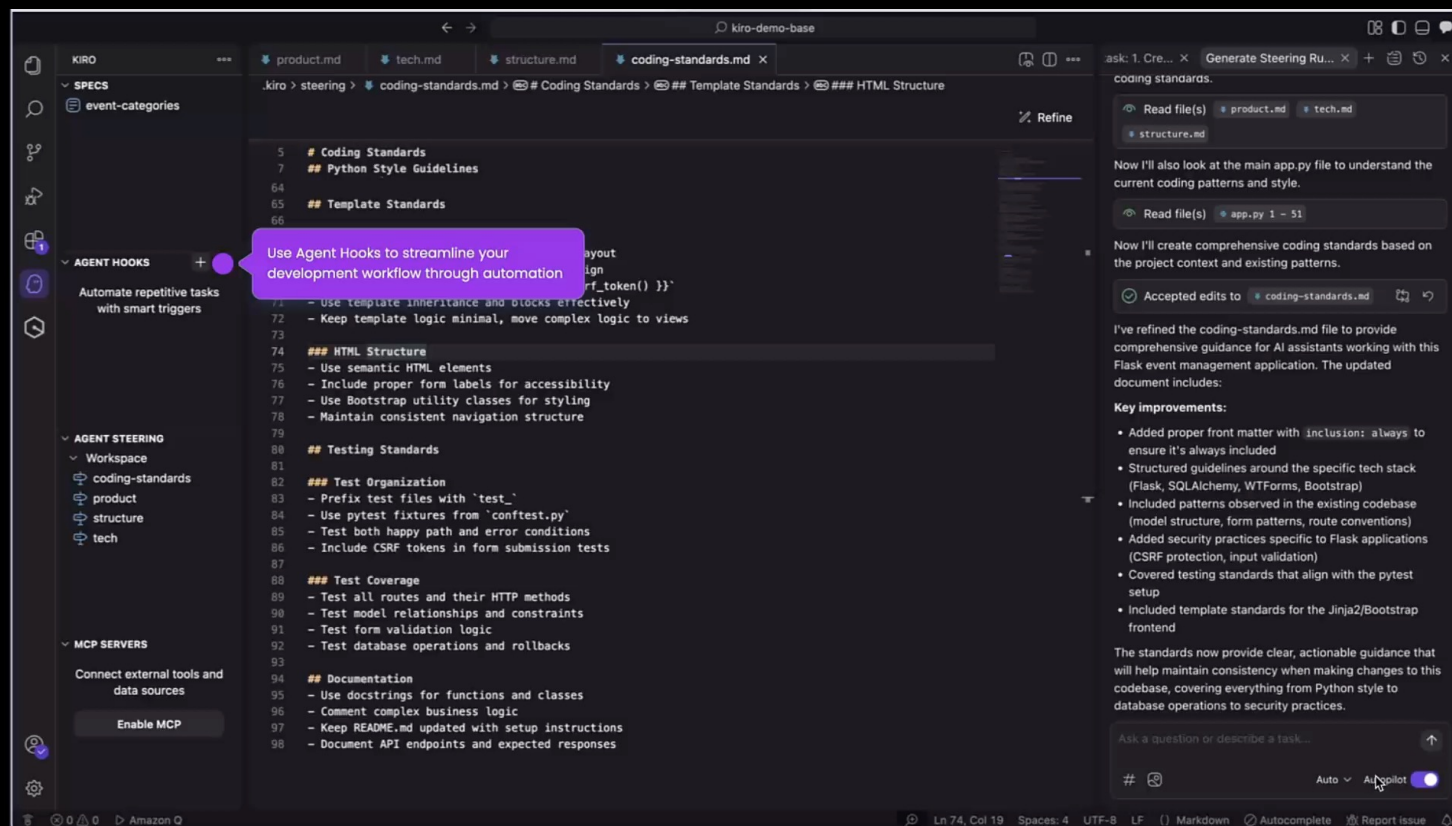
- 修改 API 程式碼後自動觸發 Agent 修改 API 文
- 修改資料庫初始化 SQL 後自動修改資料庫設計
- 修改函數後，自動檢查函數的描述

單元測試：

寫完程式碼保存後，自動補充該程式碼的單元測試

優化程式碼：

寫完程式碼後，自動觸發審核 Agent 優化程式碼



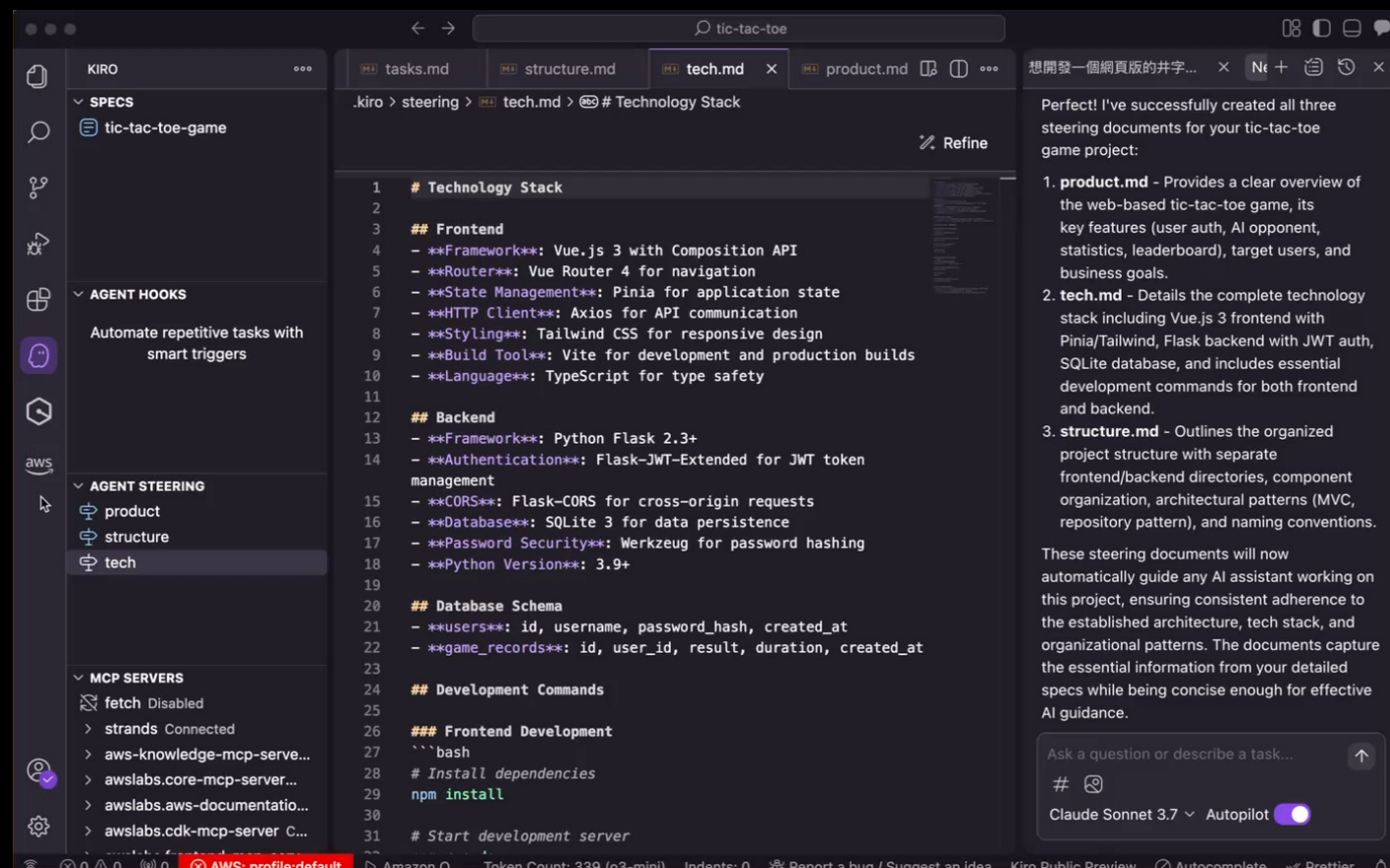
Agent steering

透過 Steering 保存永久性的知識在您專案目錄中的 `./kiro/steering/*.md` 檔。而非透過每次對話來加入上下文，來確保 KIRO 可以持續地遵循您訂立的規範

- 生成一致風格的程式碼
- 減少重複提示詞
- 訂製團隊標準
- 可擴展性的專案知識

基礎 steering file

- `product.md` – 定義業務邏輯
- `tech.md` – 定義框架、依賴庫、技術約束
- `Structure.md` – 專案結構、命名慣例



開始使用 Kiro!

kiro.dev

安裝 Kiro

<https://kiro.dev/downloads/>



CLIWEBPOWERSENTERPRISEPRICINGDOCSCOMMUNITYRESOURCES

🔍

SIGN IN

DOWNLOADS

Download Kiro

By downloading and using Kiro, you agree to the [AWS Customer Agreement](#) (or other agreement with us governing your use of AWS services), [Service Terms](#), and [Privacy Notice](#).

IDE

Download for macOS, [Linux](#), and [Windows](#).

🍏 Download IDE for macOS

CLI

Install on macOS, [Linux](#), or [Windows](#)

```
curl -fsSL https://cli.kiro.dev/install | bash
```

Latest IDE0.12.200



macOS (Apple Silicon)

⬇

macOS (Intel)

⬇

macOS (Apple Silicon, pkg)

⬇

macOS (Intel, pkg)

⬇



Windows (x64)

⬇



Linux (Debian/Ubuntu 24+)

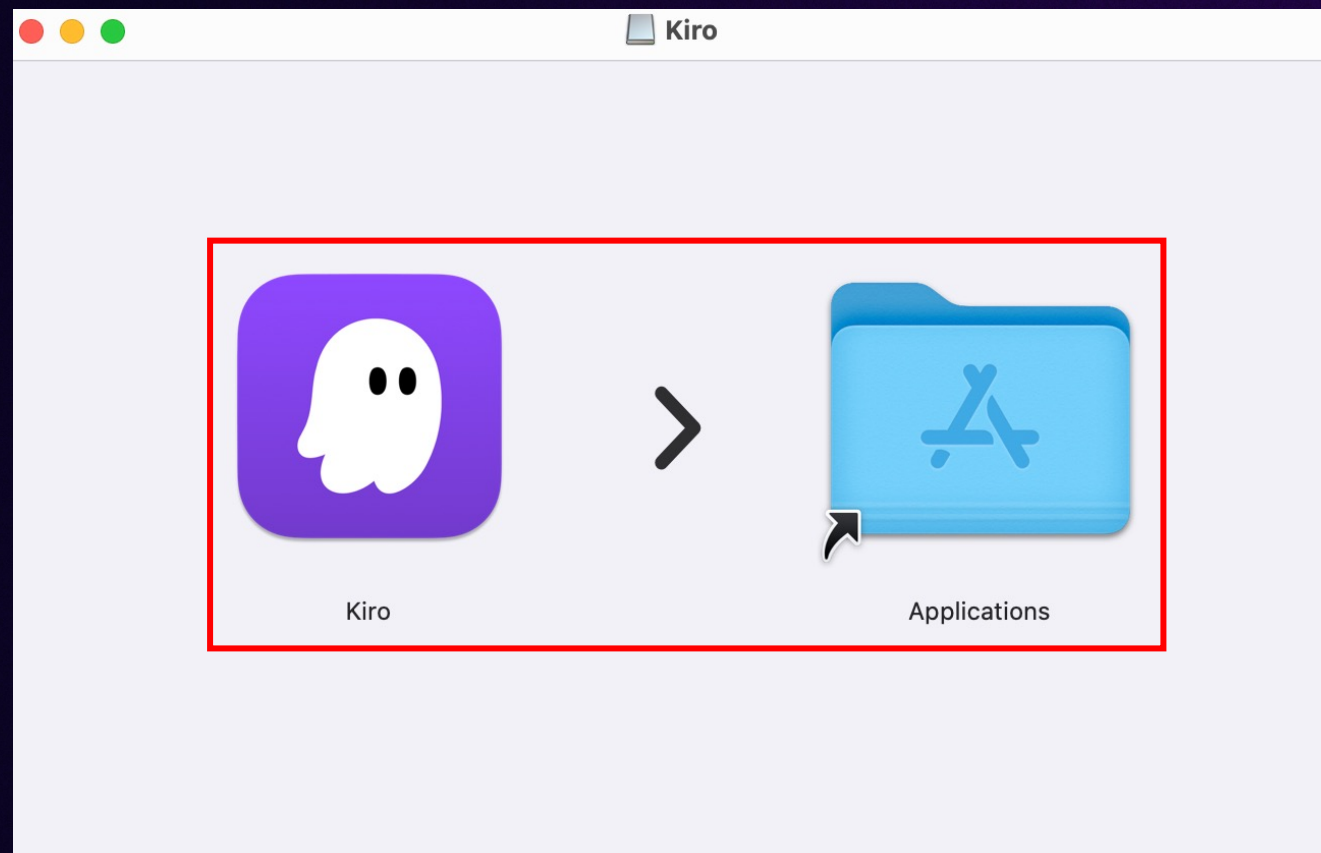
⬇

Linux (Universal)

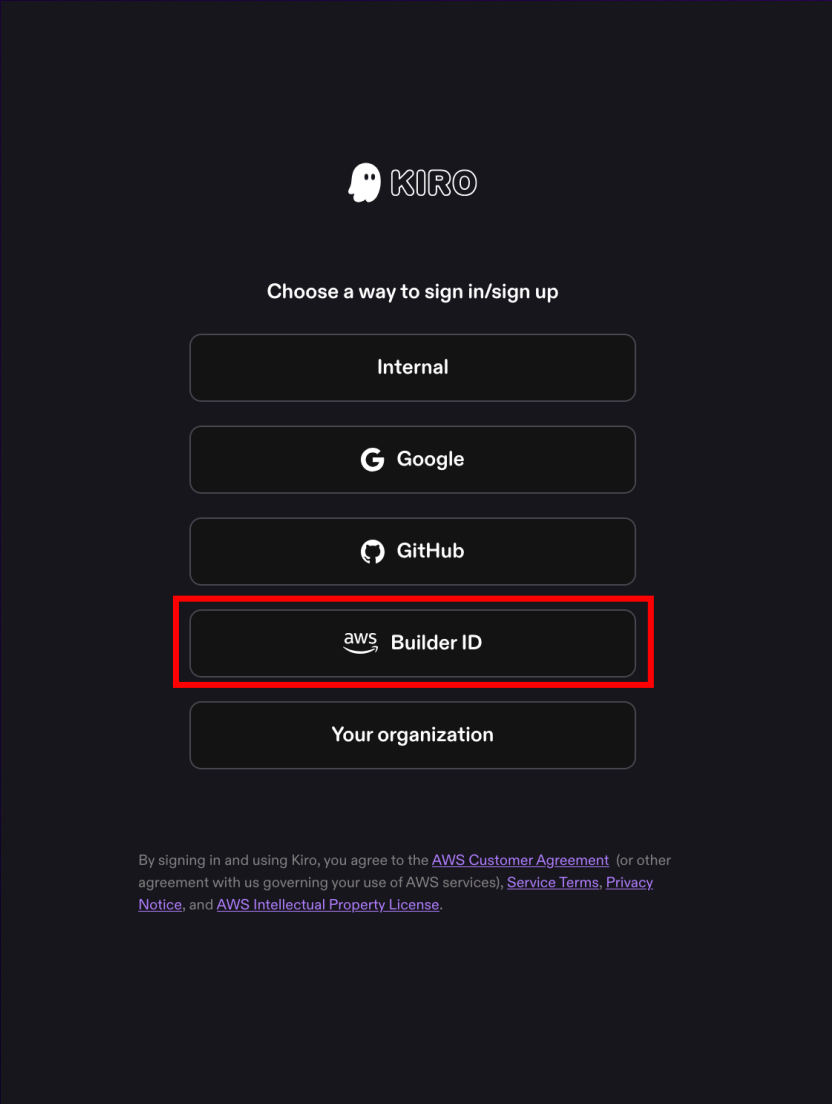
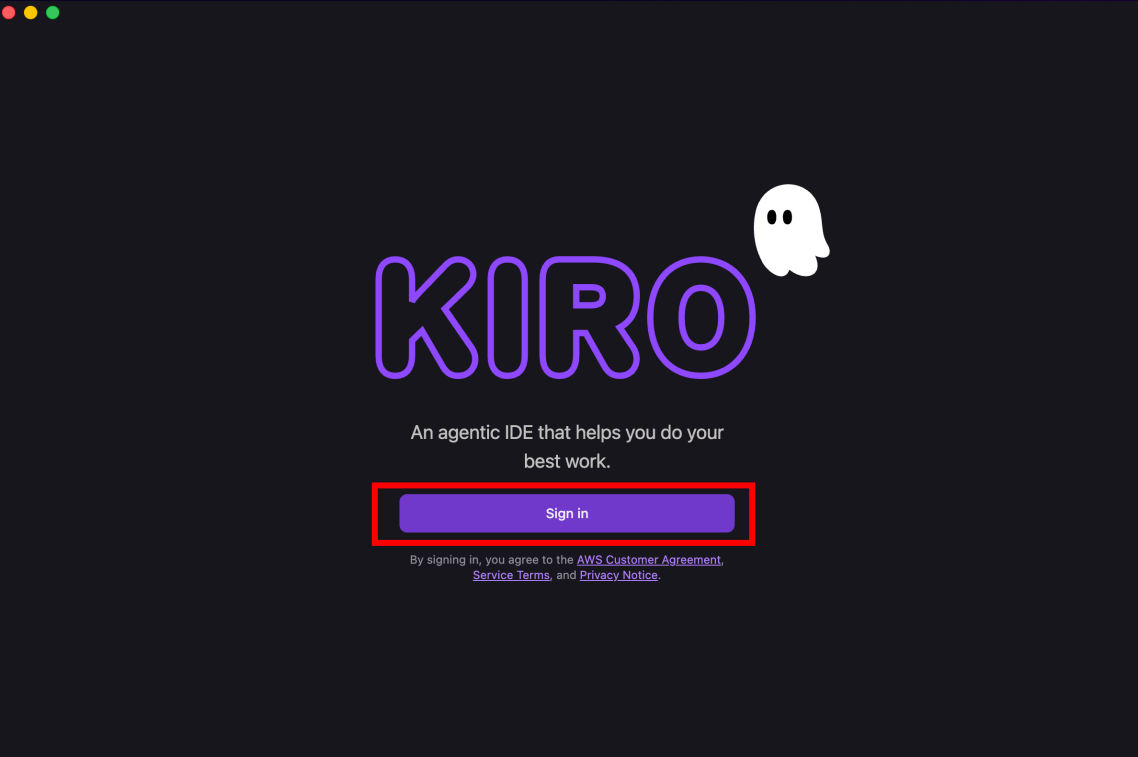
⬇



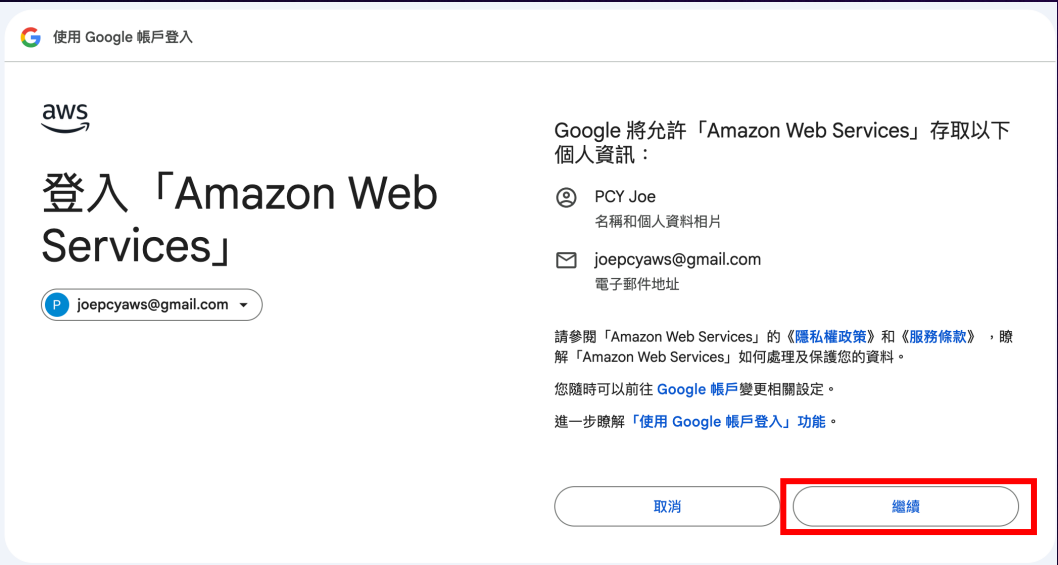
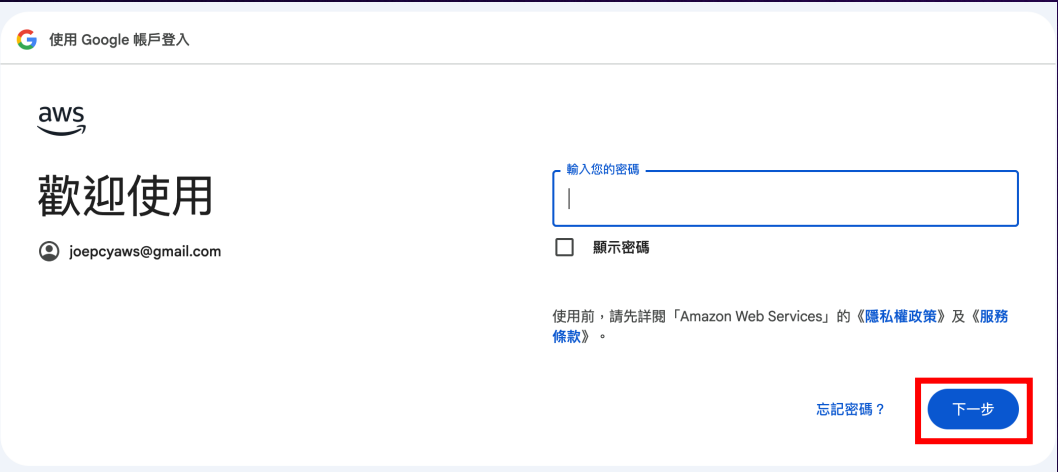
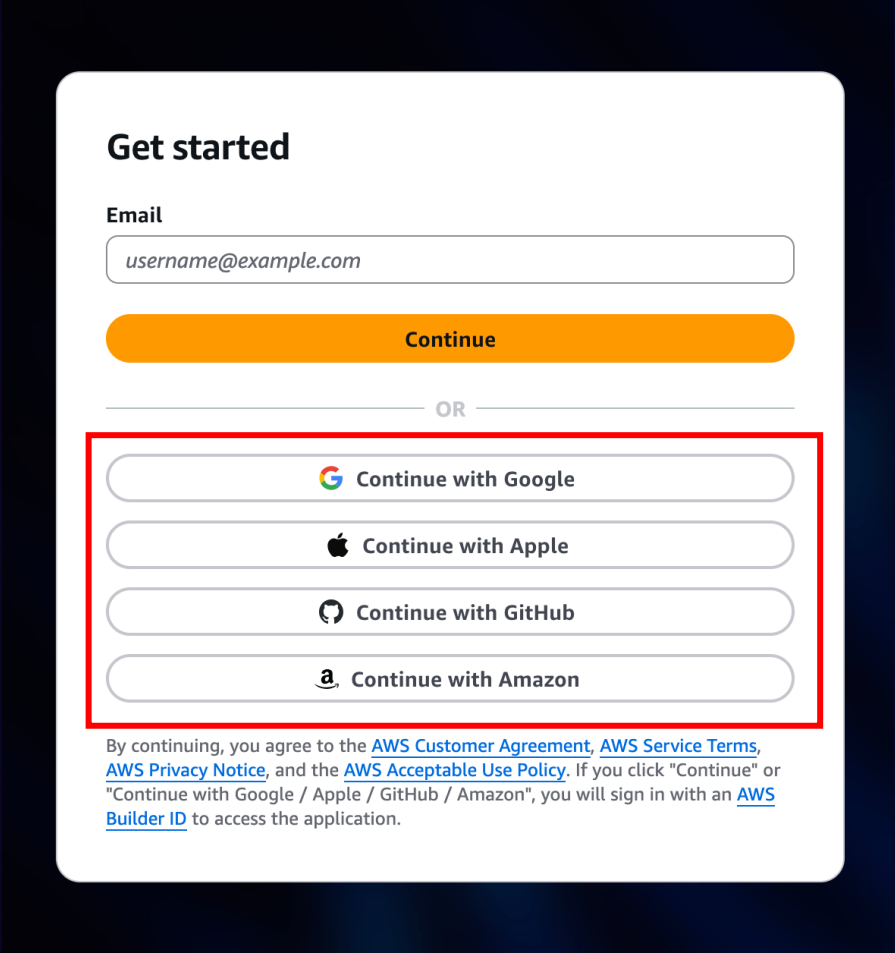
Kiro 安裝流程



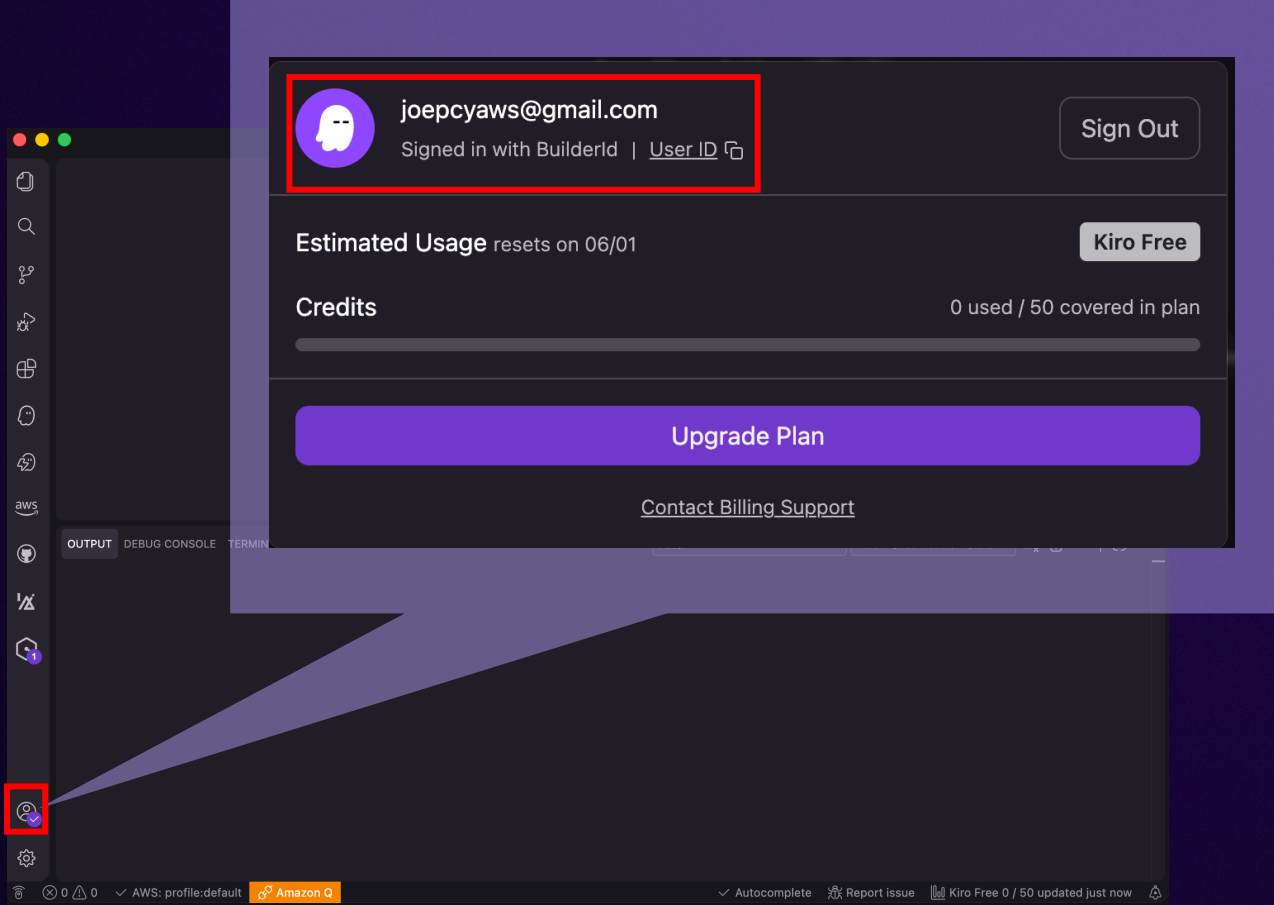
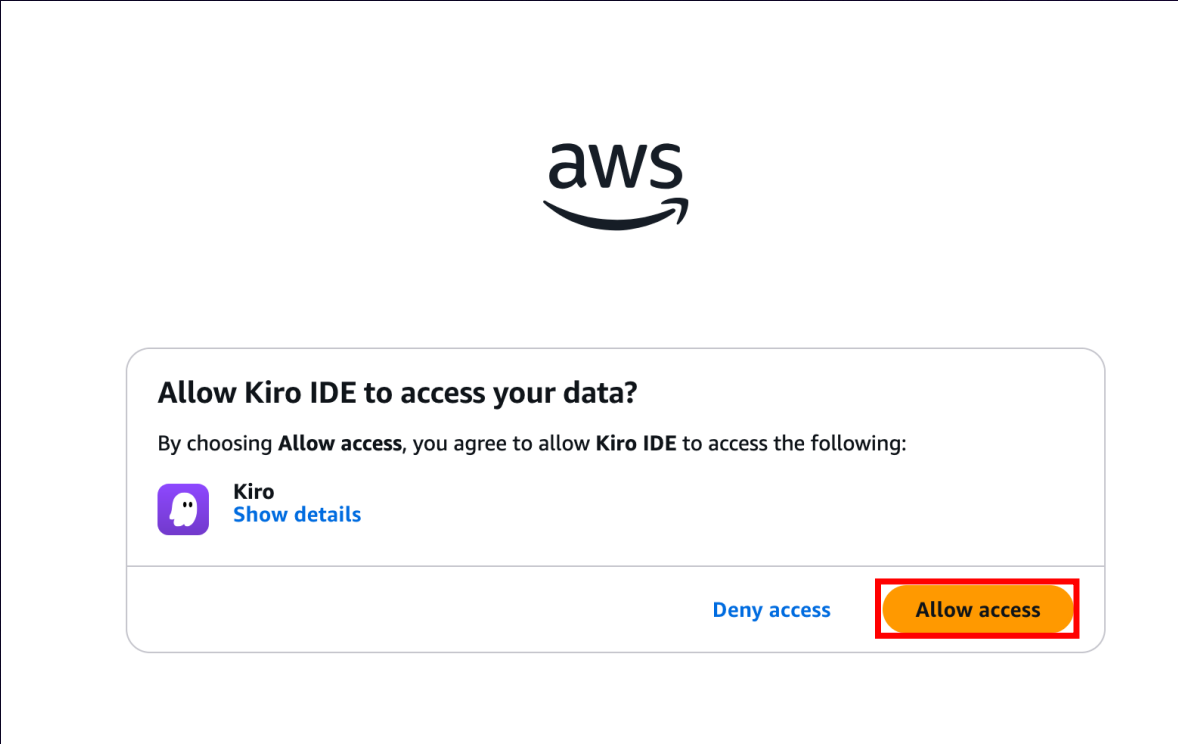
登入 Kiro IDE



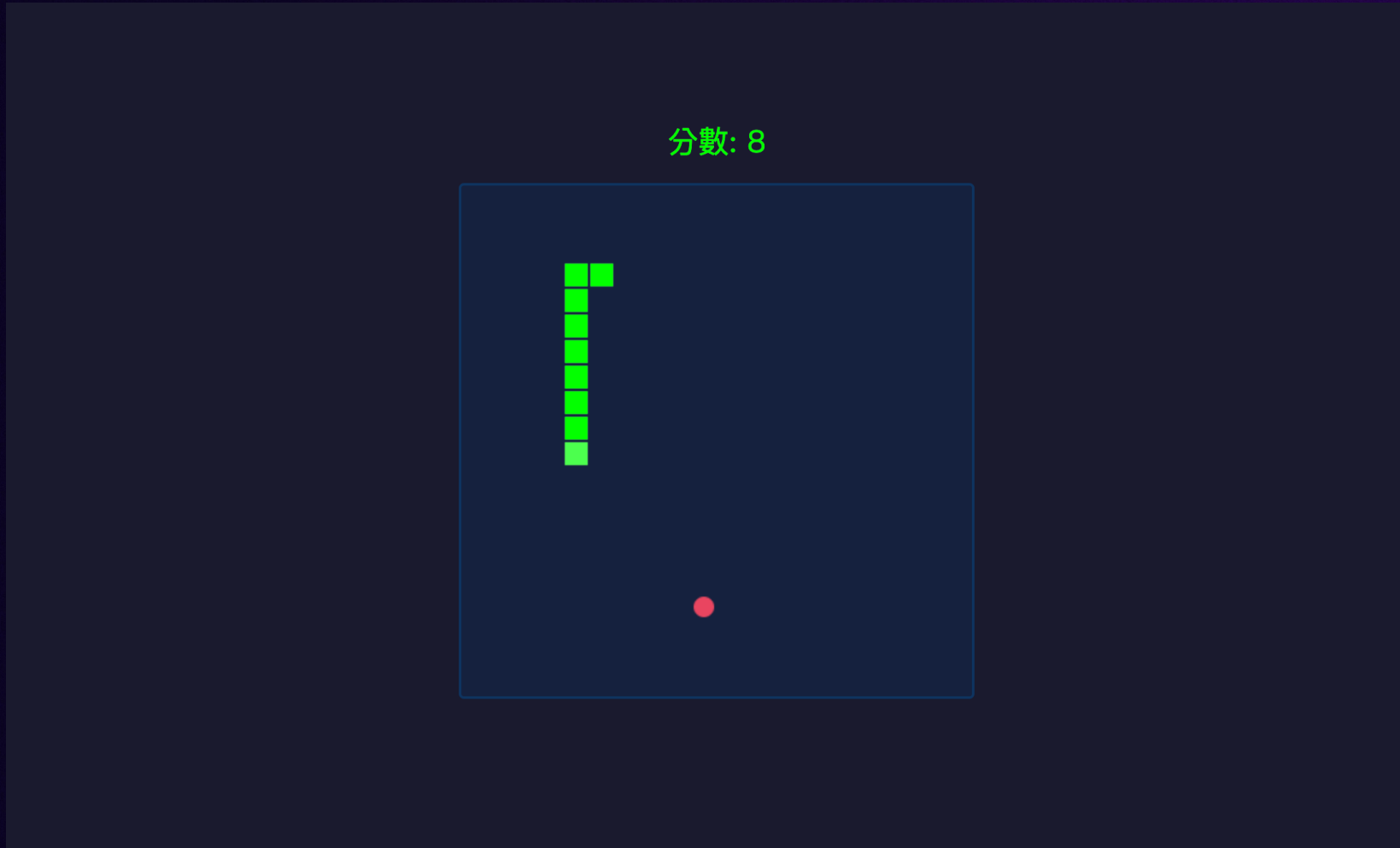
註冊 AWS Builder ID (免費)



註冊 AWS Builder ID (免費)

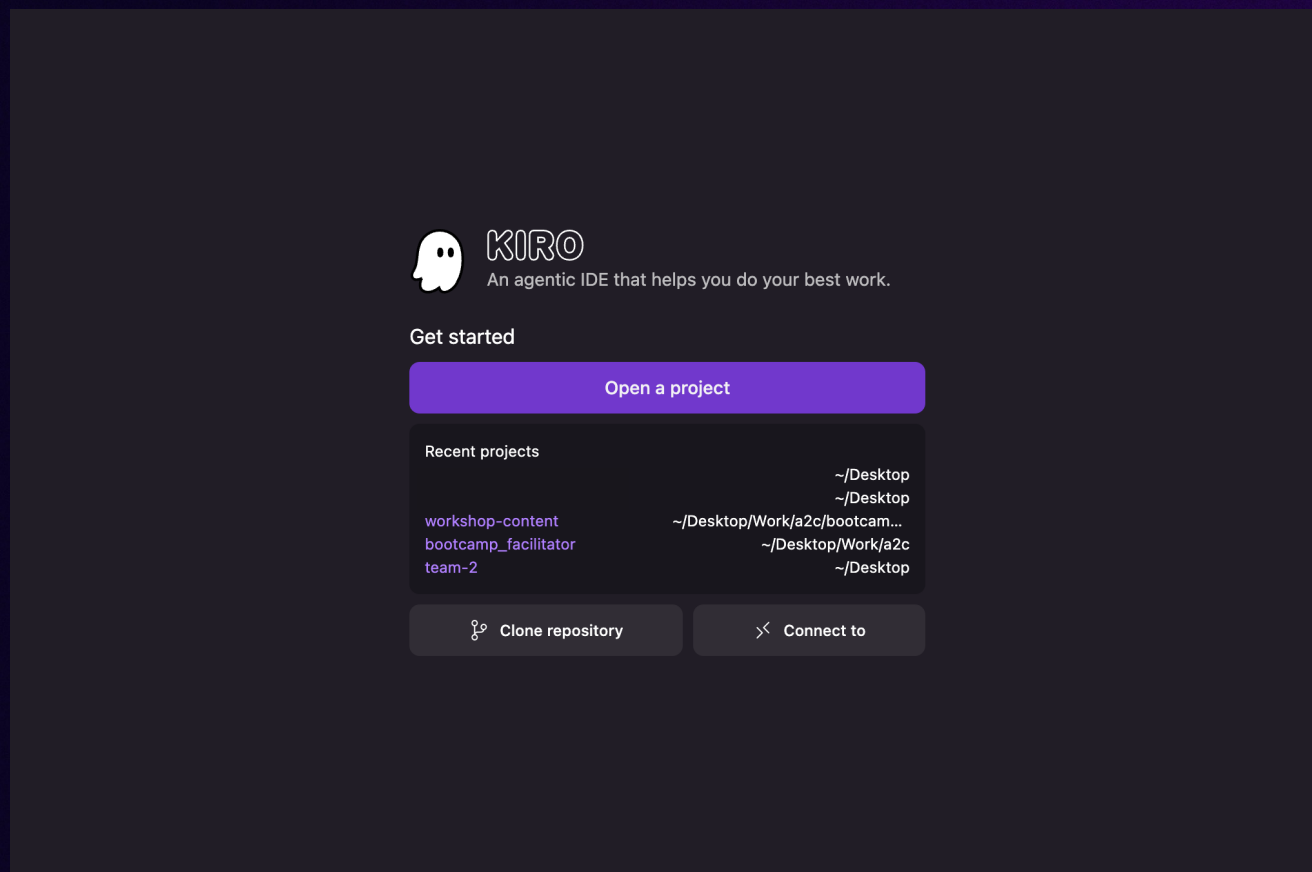


範例：貪食蛇遊戲



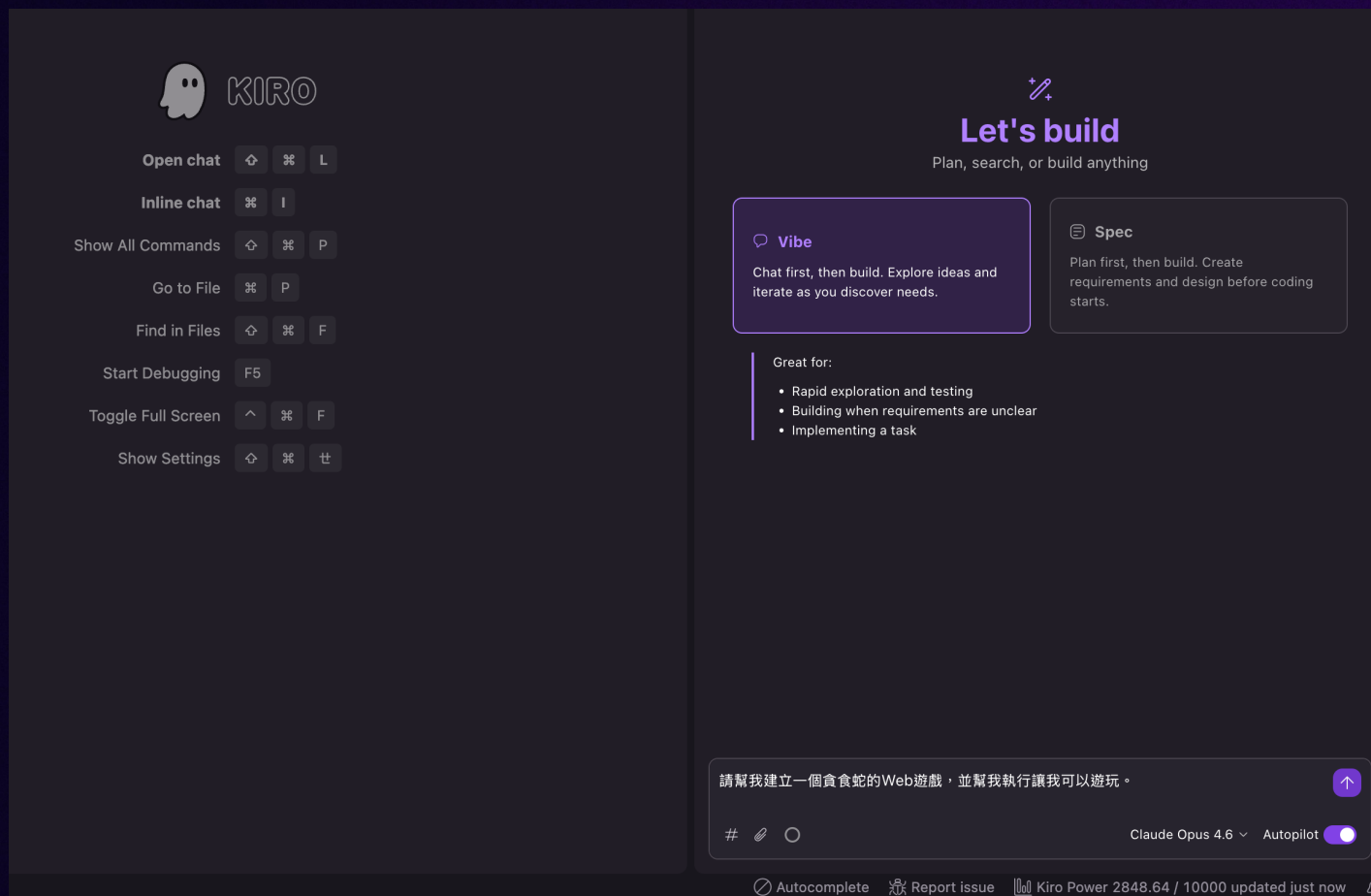
範例：貪食蛇遊戲

1. 在桌面建立“Game”資料夾，並在 Kiro 中開啟。



範例：貪食蛇遊戲

2. 在右側的聊天框中選擇 Vibe 模式，並輸入“請幫我建立一個貪食蛇的 Web 遊戲，並幫我執行讓我可以遊玩。”



範例：貪食蛇遊戲

GAME

snake.html

snake.html

```
1 <!DOCTYPE html>
2 <html lang="zh-TW">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>貪食蛇</title>
7   <style>
8     * { margin: 0; padding: 0; box-sizing: border-box; }
9     body {
10       display: flex; justify-content: center; align-items: center;
11       min-height: 100vh; background: #1a1a2e; font-family: sans-serif;
12       flex-direction: column; gap: 16px;
13     }
14     #score { color: #0ff0; font-size: 24px; }
15     canvas { background: #16213e; border: 2px solid #0f3460; border-radius:
16     #msg { color: #eee; font-size: 18px; min-height: 24px; }
17   </style>
18 </head>
19 <body>
20   <div id="score">分數: 0</div>
21   <canvas id="game" width="400" height="400"></canvas>
22   <div id="msg">按任意方向鍵開始</div>
23   <script>
24     const canvas = document.getElementById('game');
25     const ctx = canvas.getContext('2d');
26     const size = 20, cols = canvas.width / size, rows = canvas.height / size;
27     let snake, dir, food, score, running, interval;
28
29     function init() {
30       snake = [{x: 10, y: 10}];
31       dir = {x: 0, y: 0};
32       score = 0;
33       running = false;
34       placeFood();
35       updateScore();
36       document.getElementById('msg').textContent = '按任意方向鍵開始';
37       draw();
38     }
39
40     function placeFood() {
41       do {
42         food = {x: Math.floor(Math.random() * cols), y: Math.floor(Math.rand
43       } while (snake.some(s => s.x === food.x && s.y === food.y));
44     }
45
46     function updateScore() {
47       document.getElementById('score').textContent = '分數: ' + score;
48     }
49
50     function draw() {
51       ctx.clearRect(0, 0, canvas.width, canvas.height);
52       // food
53       ctx.fillStyle = '#e94560';
54       ctx.beginPath();
```

New Session

Search in conversation (⌘F)

Checkpoint Restore

請幫我建立一個貪食蛇的Web遊戲，並幫我執行讓我可以遊玩。

Kiro

Including Steering Documents amazon-builder-production-safety.md

Accepted creation of snake.html

Background process

Process 1: python3 -m http.server 8080

Read background process output Process 1: python3 -m http.server 8080

Serving HTTP on :: port 8080 (http://[::]:8080/) ...

遊戲已建立並啟動！在瀏覽器打開 <http://localhost:8080/snake.html> 即可遊玩。

操作方式：

- 方向鍵控制蛇的移動方向
- 吃到紅色食物得分
- 撞牆或撞到自己則遊戲結束
- 按 R 重新開始

Est. Credits Used: 1.57 Elapsed time: 33s

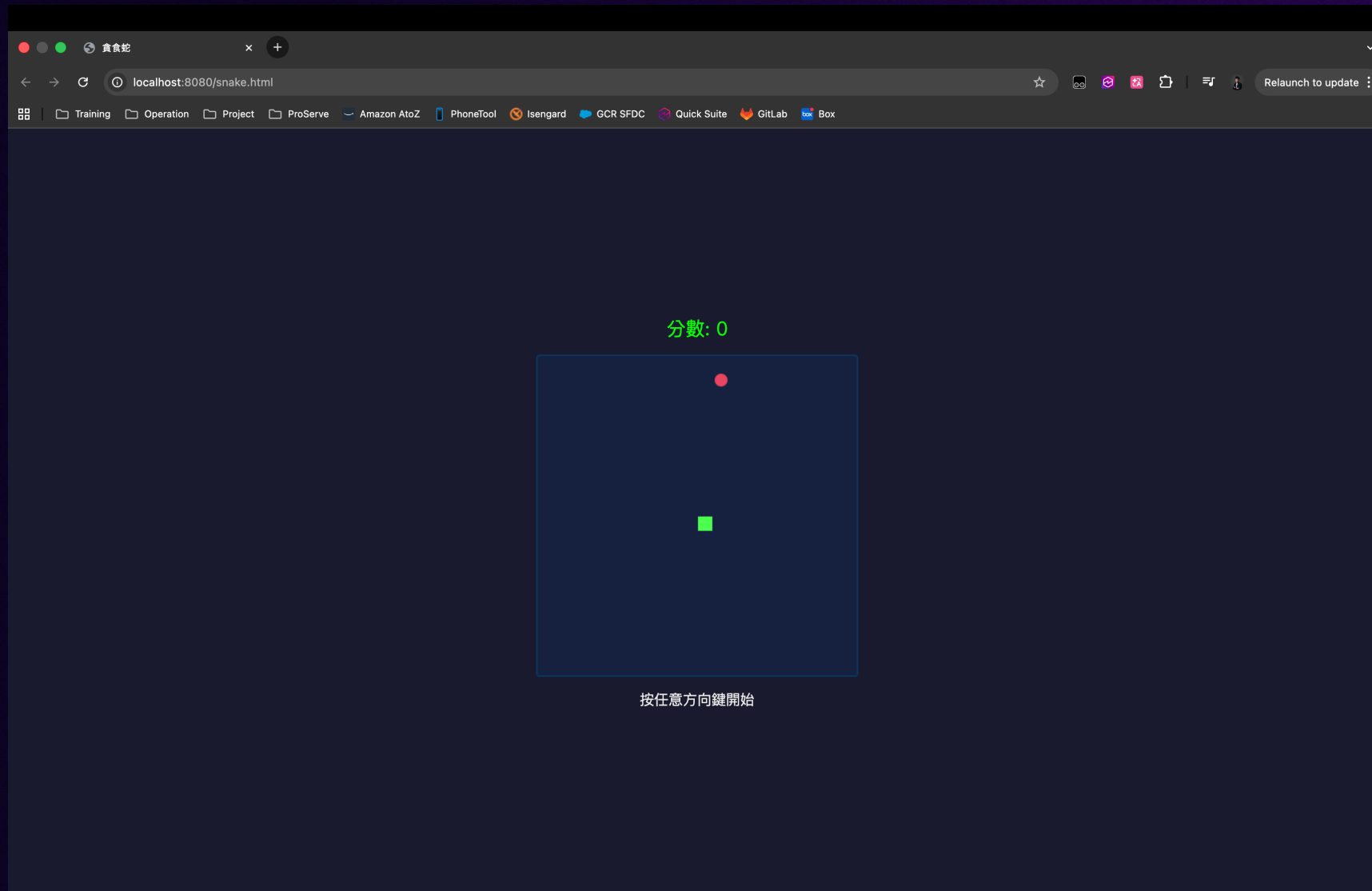
> 1 change accepted View all Revert

Ask a question or describe a task...

Claude Opus 4.6 Autopilot



範例：貪食蛇遊戲



Kiro 打造史萊姆的第一個家！

現在大家請使用 Kiro Vibe 模式打造一個可以遊玩的小遊戲，可參考以下題目：

- 圈圈叉叉
- 五子棋
- 數獨
- 象棋
- 消磚塊
- 俄羅斯方塊
- 英雄聯盟（如果可以的話XD）

Hands On Time!

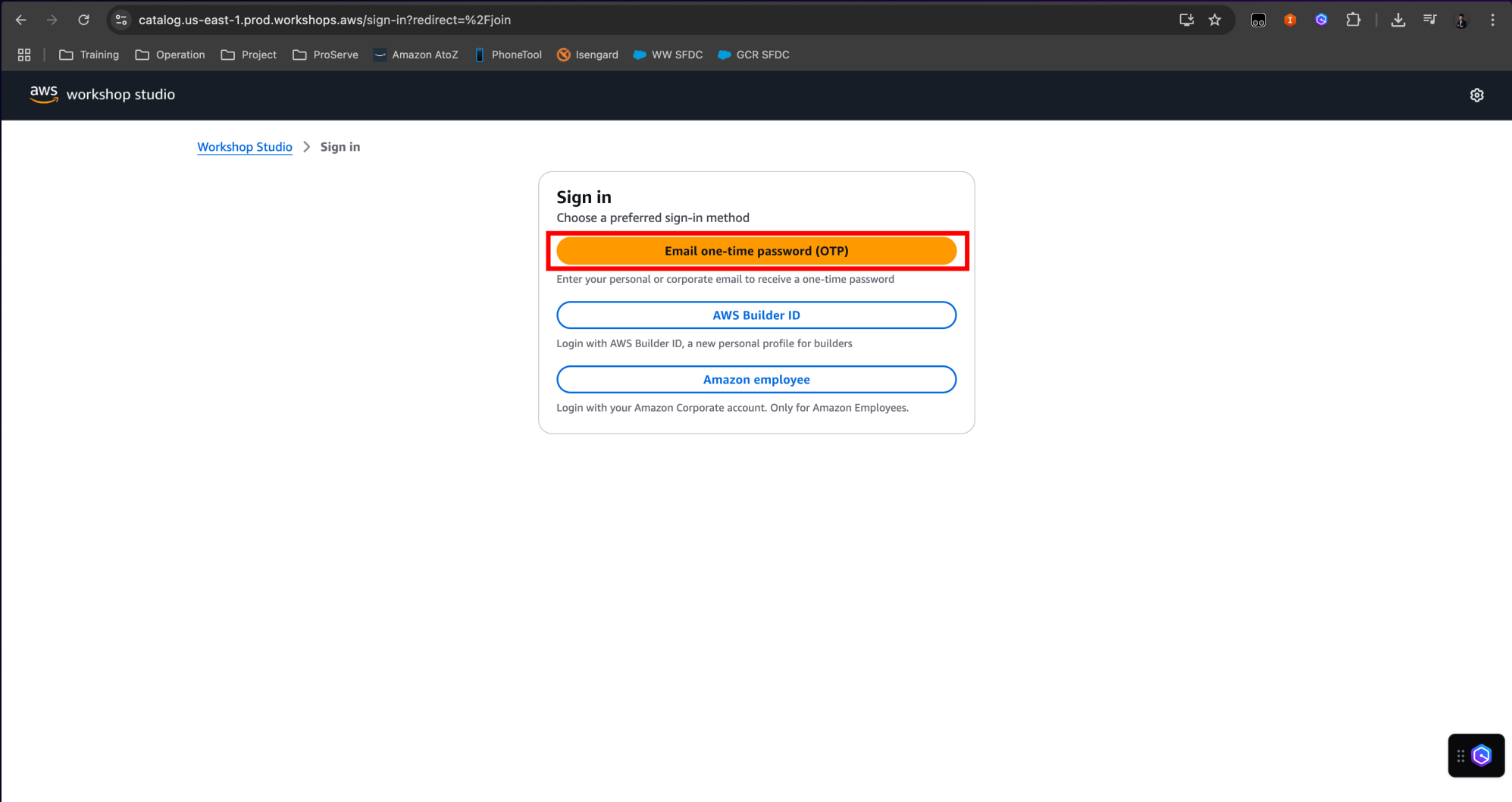


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<https://catalog.us-east-1.prod.workshops.aws/join>

Access-code: XXXXXXXXXX

Join Workshop Event



Join Workshop Event

← → ↻ login.us-east-1.auth.skillbuilder.aws/otp/input/response?redirect_uri=https%3A%2F%2Fgandalf-prod.auth.us-east-1.amazoncognito.com%2Foauth2%2Fidpresponse&state=H4slAA... ☆

Training Operation Project ProServe Amazon AtoZ PhoneTool Isengard WW SFDC GCR SFDC

aws Training and Certification English ▾

1

One-time email passcode

Send a passcode to the email below.

Email

aaronclh@amazon.com

Back

2Send passcode

[Get help signing in](#)



Join Workshop Event

Verify your AWS Training and Certification email address

N

no-reply@auth.aws.training <no-reply@auth.aws.training>

To:

Huang, Aaron

Today at 4:24 PM

aws

training and certification

To verify your email, use this one-time passcode (OTP):

1 135557546

This code will expire 5 minutes after it was sent.

AWS will never email you and ask you to disclose or verify your password, credit card, or banking account number.

Amazon Web Services, Inc. is a subsidiary of Amazon.com. Amazon.com is the registered trademark of Amazon.com, Inc. This message was produced and distributed by Amazon Web Services, Inc., 410 Terry Ave. North, Seattle, WA 98109-5210.

One-time email passcode

We sent a passcode to aaronclh@amazon.com. You should receive it within 5 minutes.

Passcode (9-digit)

Resend passcode

135557546

2

Back

Sign in

3

Get help signing in



Join Workshop Event

catalog.us-east-1.prod.workshops.aws/join

aws workshop studio

Workshop Studio > Join event

Step 1
Enter event access code

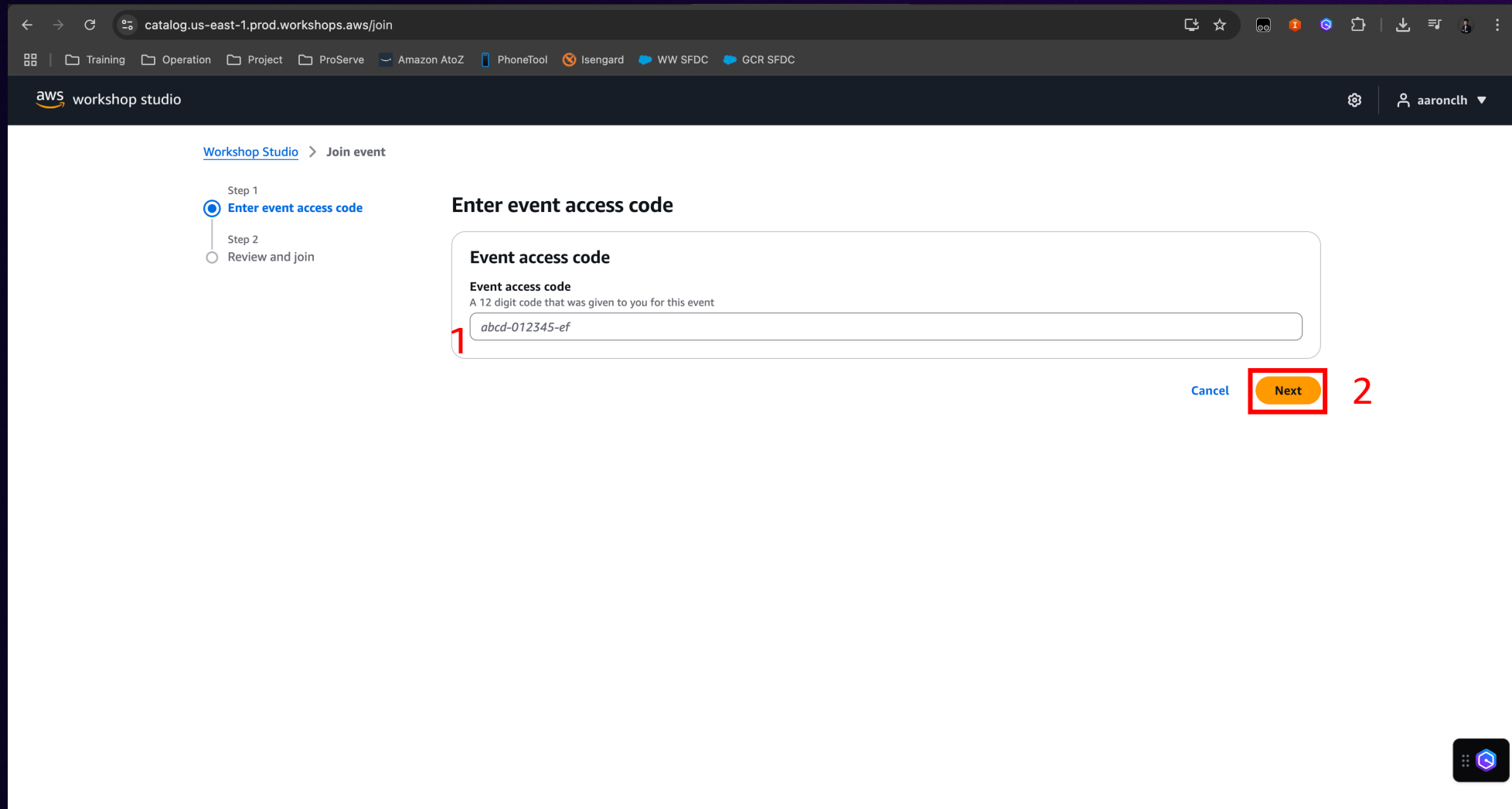
Step 2
Review and join

Enter event access code

Event access code
A 12 digit code that was given to you for this event

1

Cancel **Next** 2



Join Workshop Event

Step 1
Enter event access code

Step 2
Review and join

Event details

Name

0527 bedrock ws testing

Start time

5/27/2025 04:15 PM

Description

0527 bedrock ws testing

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Country or region

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Australia

Japan, Switzerland, Vietnam, and countries in the European Economic Area

Azerbaijan, Bolivia, Colombia, Dominican Republic, Indonesia, Lebanon, Malaysia, Mexico, Montenegro, Nepal, Philippines, Thailand, Turkey, and countries in Africa

1

☒ I agree with the Terms and Conditions

2

Join event

catalog.us-east-1.prod.workshops.aws/join

Training Operation Project ProServe Amazon AtoZ PhoneTool Isengard WW SFDC GCR SFDC

aws workshop studio

Canada, China, Republic of Korea (South Korea)

14

Australia

15

Japan, Switzerland, Vietnam, and countries in the European Economic Area

16

Azerbaijan, Bolivia, Colombia, Dominican Republic, Indonesia, Lebanon, Malaysia, Mexico, Montenegro, Nepal, Philippines, Thailand, Turkey, and countries in Africa

18

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Get AWS CLI credentials

Joe_Demo

AIWave: Taiwan Generative AI Application Hackathon

What to expect?

Schedule

Environment Setup

AWS Services

AWS CLI and SDK Access

▼ AWS account access

[Open AWS console \(us-west-2\)](#)

[Get AWS CLI credentials](#)

Share feedback

Workshop catalog in AWS Builder Center

Exit event

Event ends in 2 days 7 minutes.

[Event dashboard](#) > AIWave: Taiwan Generative AI Application Hackathon

Joe_Demo

Event information

Start time 8/05/2026 01:40 PM	Duration 72 hours	Accessible Regions us-west-2, us-east-1	Description test 8/15 env
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Workshop

Title
AIWave: Taiwan Generative AI Application Hackathon

Complexity level
400

Description
AIWave: Taiwan Generative AI Application Hackathon

Topics
Generative AI, Machine Learning (ML/AI)

AWS services
Amazon Bedrock, Amazon SageMaker

[Get started](#)

Event outputs (0)

Key	Value	Stack name	Description	Type
No stack outputs				
No stack outputs to display.				



Copy AWS CLI credentials

AWS account access

These environment variables need to be set in your terminal session to run AWS CLI commands against your temporary AWS account. Learn more about the AWS CLI in the [User Guide Version 2](#).

 **AWS CLI credentials**

Linux or macOS (bash)

Linux or macOS (fish)

Windows (PowerShell)

Windows (cmd)

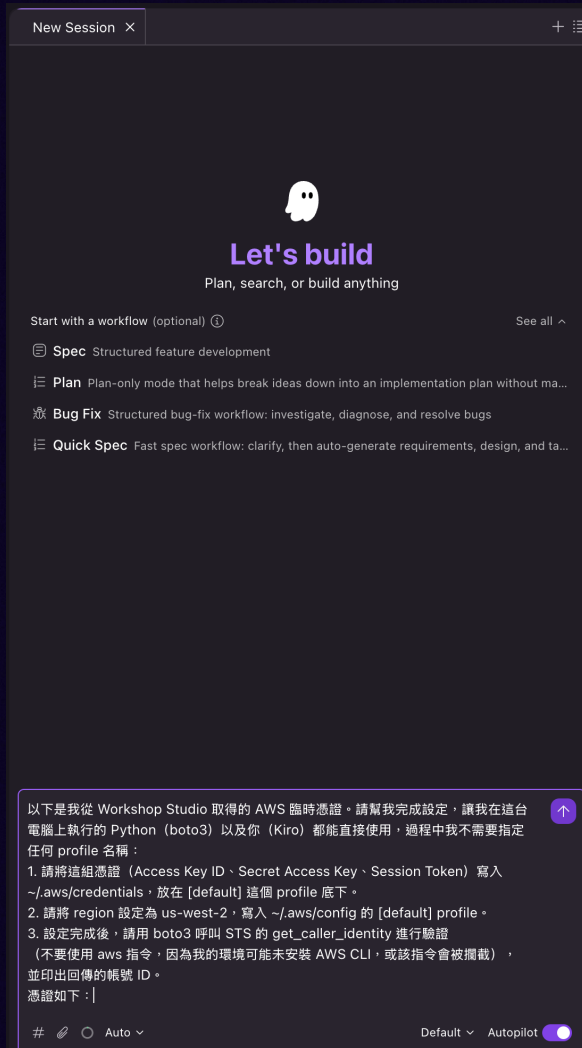
```
export AWS_DEFAULT_REGION="us-west-2"
export AWS_ACCESS_KEY_ID=[REDACTED]
export AWS_SECRET_ACCESS_KEY=[REDACTED]
export AWS_SESSION_TOKEN=[REDACTED]
```



 Only use `us-west-2` and `us-east-1` as your Region, unless otherwise directed by the event operator. You can optionally set the `AWS_DEFAULT_REGION` environment variable to one of the allowed Regions to set the default Region for CLI commands. This value can be overridden with the `--region` command line parameter.

Close

Config AWS CLI credentials to Kiro



以下是我從 Workshop Studio 取得的 AWS 臨時憑證。請幫我完成設定，讓我在這台電腦上執行的 Python (boto3) 以及你 (Kiro) 都能直接使用，過程中我不需要指定任何 profile 名稱：

1. 請將這組憑證 (Access Key ID、Secret Access Key、Session Token) 寫入 `~/.aws/credentials`，放在 `[default]` 這個 profile 底下。
2. 請將 region 設定為 `us-west-2`，寫入 `~/.aws/config` 的 `[default]` profile。
3. 設定完成後，請用 `boto3` 呼叫 STS 的 `get_caller_identity` 進行驗證 (不要使用 `aws` 指令，因為我的環境可能未安裝 AWS CLI，或該指令會被攔截)，並印出回傳的帳號 ID。

憑證如下：

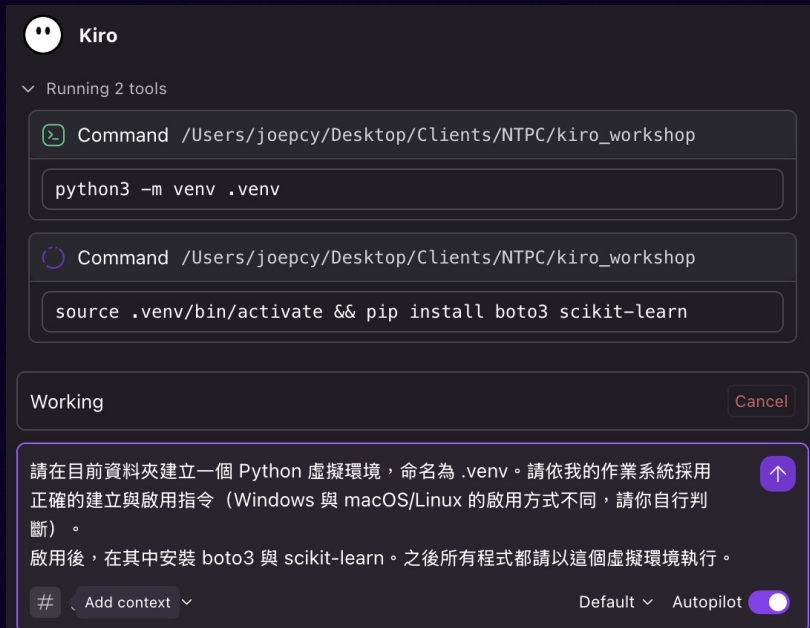
[icon] Auto ▾

Default ▾ Autopilot ☒

Check CLI credentials status

```
> python3 -c "  
dquote> import boto3  
dquote> sts = boto3.client('sts')  
dquote> identity = sts.get_caller_identity()  
dquote> print(f\"Account ID: {identity['Account']}\")  
dquote> print(f\"ARN: {identity['Arn']}\")  
dquote> "  
Account ID: 09[REDACTED]94  
> aws sts get-caller-identity  
{  
    "UserId": "AROARNK3NJEROQHLMFFAN:Participant",  
    "Account": "09[REDACTED]94",  
    "Arn": "arn:aws:sts::097365346594:assumed-role/WSParticipantRole/Participant"  
}
```

Create venv and install boto3 & scikit-learn



請在目前資料夾建立一個 Python 虛擬環境，命名為 `.venv`。請依我的作業系統採用正確的建立與啟用指令（Windows 與 macOS/Linux 的啟用方式不同，請你自行判斷）。

啟用後，在其中安裝 `boto3` 與 `scikit-learn`。之後所有程式都請以這個虛擬環境執行。

Auto ▾

Default ▾ Autopilot

Prepare training data for SageMaker

 Kiro

> 3 tool calls

完成了。

- 虛擬環境已建立於 `.venv` (Python 3.12)
- 已安裝 `boto3 1.43.65` 和 `scikit-learn 1.9.0` (含所有相依套件)

macOS 上的啟用方式是 `source .venv/bin/activate`。後續所有程式我都會透過 `.venv/bin/python3` 來執行，確保使用這個虛擬環境。

Est. Credits Used: 0.44 Elapsed time: 2m 33s

我的帳號目前是空的，region 為 us-west-2。我要使用 SageMaker 內建的 XGBoost 演算法建立一個表格分類模型。請協助我：

1. 使用 scikit-learn 的 Iris 資料集產生約 120 筆訓練資料。
2. 寫成符合 SageMaker XGBoost 要求的 CSV 格式：不含表頭、第一欄為 label (數字)，存成 train.csv。
3. 使用 boto3 建立一個新的 S3 bucket (名稱取易於辨識者，例如 kiro-workshop-<你的名字>)。請注意 us-west-2 建立 bucket 需指定 LocationConstraint。
4. 將 train.csv 上傳至該 bucket。
5. 印出資料在 S3 的完整路徑 (s3://...)。

ⓘ 🔴 Auto ▾ Default ▾ Autopilot ☒

我的帳號目前是空的，region 為 us-west-2。我要使用 SageMaker 內建的 XGBoost 演算法建立一個表格分類模型。請協助我：

1. 使用 scikit-learn 的 Iris 資料集產生約 120 筆訓練資料。
2. 寫成符合 SageMaker XGBoost 要求的 CSV 格式：不含表頭、第一欄為 label (數字)，存成 train.csv。
3. 使用 boto3 建立一個新的 S3 bucket (名稱取易於辨識者，例如 kiro-workshop-<你的名字>)。請注意 us-west-2 建立 bucket 需指定 LocationConstraint。
4. 將 train.csv 上傳至該 bucket。
5. 印出資料在 S3 的完整路徑 (s3://...)。

ⓘ 🔴 Auto ▾

Default ▾ Autopilot ☒

Check training data for SageMaker

Amazon S3

Buckets

General purpose bucket

Directory buckets

Table buckets

Vector buckets

Files

Access management and security

Buckets

General purpose buckets (1) Info

Buckets are containers for data stored in S3.

Find buckets by name

Name

AWS Region

Creation date

kiro-workshop-ntpc-iris

US West (Oregon) us-west-2

August 6, 2026, 13:55:47 (UTC+08:00)

Amazon S3

Buckets

General purpose buckets

Directory buckets

Table buckets

Vector buckets

Files

Access management and security

train/

Objects (1/1)

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

☒

Name

Type

Last modified

Size

Storage class

☒

train.csv

csv

August 6, 2026, 13:55:47 (UTC+08:00)

2.2 KB

Standard



Create training job in SageMaker

1. 從 Iris 取 120 筆訓練資料
2. 存成 SageMaker XGBoost CSV 格式的 `train.csv`
3. 建立 S3 bucket `kiro-workshop-ntpc-iris` (含 LocationConstraint)
4. 上傳 `train.csv`
5. 印出完整 S3 路徑

如果 bucket 名稱已被佔用 (全球唯一)，請告訴我，我會換一個名稱。

Est. Credits Used: 0.09 Elapsed time: 7s

請使用 boto3 (不要使用 sagemaker SDK)，於 `.venv` 中撰寫一個 `train.py`，呼叫 SageMaker 的 `create_training_job`，使用內建的 XGBoost 演算法訓練我剛才上傳的 `train.csv`。

關於 execution role，請依下列邏輯處理：

1. 使用 boto3 列出帳號中的 IAM roles，尋找可用的 SageMaker 執行角色。
2. 必須過濾掉路徑含有 `aws-service-role` 的角色 (此為 service-linked role，不能作為訓練任務的 RoleArn)。
3. 若找不到可用角色，請自動建立一個：名稱為 `kiro-sagemaker-execution-role`，信任政策允許 `sagemaker.amazonaws.com` 擔任，附加 `AmazonSageMakerFullAccess`

📎 ○ Auto ▾

Default ▾ Autopilot ☒

請使用 boto3 (不要使用 sagemaker SDK)，於 `.venv` 中撰寫一個 `train.py`，呼叫 SageMaker 的 `create_training_job`，使用內建的 XGBoost 演算法訓練我剛才上傳的 `train.csv`。

關於 execution role，請依下列邏輯處理：

1. 使用 boto3 列出帳號中的 IAM roles，尋找可用的 SageMaker 執行角色。
2. 必須過濾掉路徑含有 `aws-service-role` 的角色 (此為 service-linked role，不能作為訓練任務的 RoleArn)。
3. 若找不到可用角色，請自動建立一個：名稱為 `kiro-sagemaker-execution-role`，信任政策允許 `sagemaker.amazonaws.com` 擔任，附加 `AmazonSageMakerFullAccess`

📎 ○ Auto ▾

Default ▾ Autopilot ☒

Check training job in SageMaker

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Canvas

RStudio

Notebooks

TensorBoard

Profiler

Partner AI Apps

▼ Model training & customization

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New

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kiro-xgb-ntpc-20260806-060026

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New

Nova Forge

New

HyperPod clusters

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kiro-xgb-ntpc-20260806-060026

Clone

Create model package

Stop

Create model

Job settings

Job name

kiro-xgb-ntpc-20260806-060026

Status

Completed

View history

SageMaker metrics time series

Disabled

IAM role ARN

arn:aws:iam::097365346594:role/kiro-sagemaker-execution-role

ARN

arn:aws:sagemaker:us-west-2:097365346594:training-job/kiro-xgb-ntpc-20260806-060026

Creation time

Aug 06, 2026 06:00 UTC

Last modified time

Aug 06, 2026 06:02 UTC

Training time (seconds)

100

Billable time (seconds)

100

Managed spot training savings

0%

Tuning job source/parent

-

Algorithm

Algorithm ARN

-

Additional volume size (GB)

10

Maximum wait time for managed spot trainings

-

Volume encryption key

-

Training image

246618743249.dkr.ecr.us-west-2.amazonaws.com/sagemaker-xgboost:1.7-1

Maximum runtime (seconds)

3600

Managed spot training

Disabled

Input mode

File

Instance group

Instance type

Instance count

Keep alive period



Create real-time endpoint in sagemaker

1. 列出 IAM roles，找到信任 `sagemaker.amazonaws.com` 且路徑不含 `aws-service-role` 的角色。找不到就自動建立 `kiro-sagemaker-execution-role`（附加 `SageMakerFullAccess` + `S3FullAccess`），等 10 秒讓 IAM 生效。
2. 使用 `us-west-2` 的 XGBoost 1.7-1 內建容器映像。
3. 呼叫 `create_training_job`，訓練資料指向 `s3://kiro-workshop-ntpc-iris/sagemaker/xgboost-iris/train`，instance 為 `ml.m5.xlarge` x 1。
4. 超參數：`multi:softmax`、`num_class=3`、`num_round=100`。
5. 模型輸出至同一 bucket 的 `output` 路徑。
6. 印出訓練任務名稱。

Est. Credits Used: 0.69 Elapsed time: 3m 42s

1 file changed

View changes (1)

Revert changes (1)

請使用 boto3 將我剛才訓練完成的 XGBoost 模型（`model.tar.gz` 位於 S3）部署為一個 real-time endpoint。步驟為 `create_model`、`create_endpoint_config`、`create_endpoint`。

- 自動取得最近一次訓練任務的 model artifact 路徑與其使用的 role ARN。
- 使用相同的 XGBoost 容器 image URI（推論亦使用此容器）。
- 推論 instance 使用 `ml.m5.xlarge`，一台。
- endpoint 命名為 `kiro-xgb-endpoint-<你的名字>`。
- 使用 `waiter` 等待 endpoint 變為 `InService`（約 5 至 8 分鐘），並印出 endpoint 名稱。

撰寫成 `deploy.py` 並執行。

Auto

Default

Autopilot

請使用 boto3 將我剛才訓練完成的 XGBoost 模型（`model.tar.gz` 位於 S3）部署為一個 real-time endpoint。步驟為 `create_model`、`create_endpoint_config`、`create_endpoint`。

- 自動取得最近一次訓練任務的 model artifact 路徑與其使用的 role ARN。
- 使用相同的 XGBoost 容器 image URI（推論亦使用此容器）。
- 推論 instance 使用 `ml.m5.xlarge`，一台。
- endpoint 命名為 `kiro-xgb-endpoint-<你的名字>`。
- 使用 `waiter` 等待 endpoint 變為 `InService`（約 5 至 8 分鐘），並印出 endpoint 名稱。

撰寫成 `deploy.py` 並執行。

Check real-time endpoint in sagemaker

Amazon SageMaker AI > Endpoint

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▶ Training plans

▼ Deployments & inference

Deployable models

Endpoints

Endpoint configurations

Batch transform jobs

Shadow tests

Compilation jobs

Inference Recommender

HyperPod inference

▶ Model governance

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kiro-xgb-endpoint-ntpc

View in SageMaker Insights

Delete

Endpoint summary

Name

kiro-xgb-endpoint-ntpc

ARN

arn:aws:sagemaker:us-west-2:097365346594:endpoint/kiro-xgb-endpoint-ntpc

URL

https://runtime.sagemaker.us-west-2.amazonaws.com/endpoints/kiro-xgb-endpoint-ntpc/invocations

Learn more about the API

Alarms

0 alarms

Status

✔ InService

Creation time

Thu Aug 06 2026 14:18:40 GMT+0800 (Taiwan Standard Time)

GPU availability

0/0 GPUs available

Type

Real-time

Last updated

Thu Aug 06 2026 14:21:50 GMT+0800 (Taiwan Standard Time)

Model container logs

/aws/sagemaker/endpoints/kiro-xgb-endpoint-ntpc



Test real-time endpoint

請撰寫 `invoke.py`，使用 `boto3` 呼叫我剛部署的 endpoint
(`kiro-xgb-endpoint-<你的名字>`)。送入一筆測試特徵（四個數字，Iris 格式，
CSV 一行，不含 label），呼叫時 `ContentType` 請設定為 `text/csv`
(XGBoost endpoint 僅接受 `text/csv`)。並將回傳的類別對照 Iris 類別名稱
(0 為 `setosa`、1 為 `versicolor`、2 為 `virginica`) 後印出。完成後直接執行。



Auto ▾

Default ▾ Autopilot ☒

```
Sending to endpoint: kiro-xgb-endpoint-ntpc
Input features (CSV): 6.7,3.0,5.2,2.3

Predicted class: 2 (virginica)
```

Clean up the resource

text

Predicted class: 2 (virginica)

Est. Credits Used: 0.47 Elapsed time: 24s

1 file changed View changes (1) Revert changes (1)

請使用 boto3 刪除我剛才建立的 SageMaker endpoint (kiro-xgb-endpoint-<你的名字>)，
連同其 endpoint config 與 model 一併清除，並確認皆已刪除。

Auto Default Autopilot

請使用 boto3 刪除我剛才建立的 SageMaker endpoint (kiro-xgb-endpoint-<你的名字>)，
連同其 endpoint config 與 model 一併清除，並確認皆已刪除。

Auto Default Autopilot

Endpoints

View in SageMaker Insights

Connect to your observability tool

Update endpoint

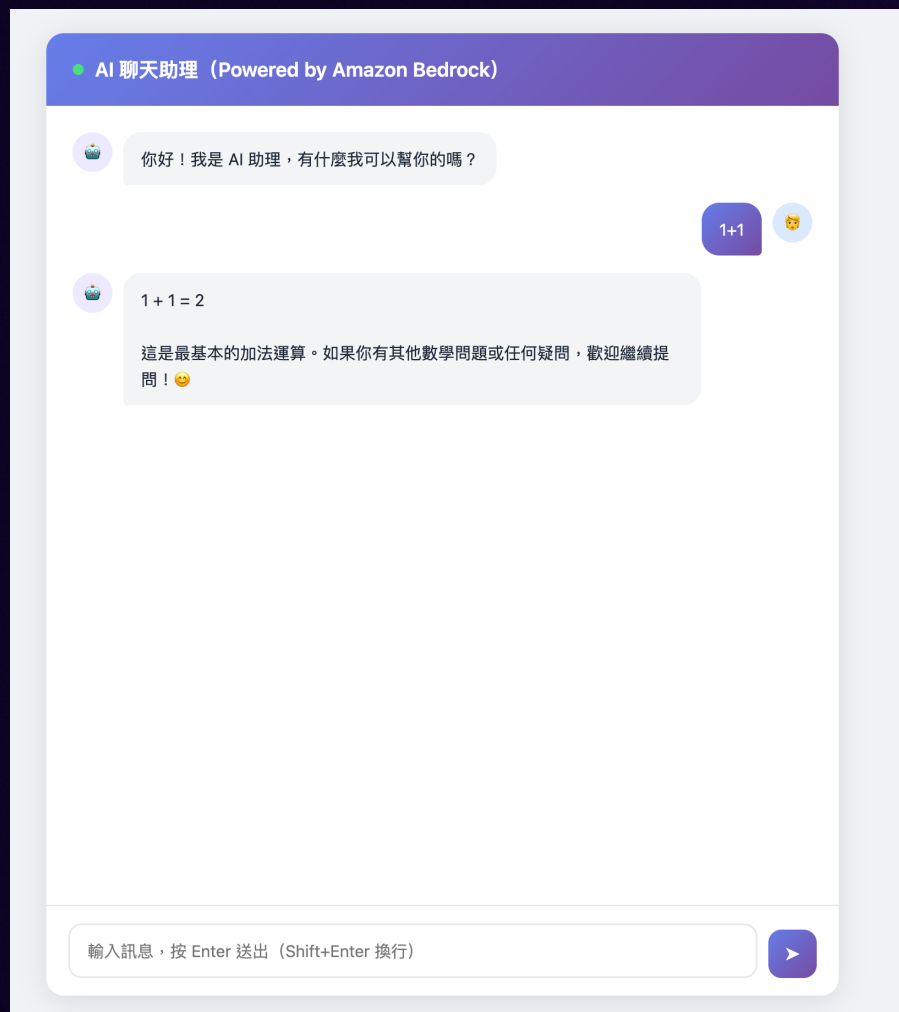
Actions

Create endpoint

Find endpoints

Name	ARN	Creation time	Status	Detailed observability	Last updated
There are currently no resources.					

在 AWS 建立聊天機器人網站範例 prompt



幫我在 AWS 上建立一個 AI 聊天機器人，需求如下：

前端：靜態網頁，部署在 S3 + CloudFront，要有漂亮的聊天介面（對話泡泡、打字動畫）

後端：API Gateway + Lambda，串接 Amazon Bedrock 的 Claude 4.5 模型

部署方式：用 Python 腳本（boto3）直接部署，不要用 CDK 或 CloudFormation

環境：用我的 profile [剛剛建立的 profile 名稱]

其他：CORS 要設定好、對話要保留歷史紀錄、最後輸出可以直接測試的網頁 URL

請直接產出所有程式碼並執行部署。